

# Lag-Aware Multi-City Air Pollution Exposure Modeling for Predicting Respiratory Hospital Admissions Using Spatial and Industrial Features

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**Abstract**—Urban air pollution constitutes one of the foremost environmental threats to human health, with its burden falling most acutely on the respiratory system. Yet most machine learning models built to forecast respiratory hospital admissions still treat pollution exposure and clinical response as simultaneous events, overlooking the multi-day biological delay between inhalation and symptom manifestation. City-level geographic heterogeneity and the modifying influence of local terrain or nearby industrial activity receive similarly sparse treatment. This paper presents a lag-aware predictive architecture evaluated across approximately 300 Indian cities for the period 2020–2025. The model fuses seven-day histories of AQI and constituent pollutant concentrations with terrain variables derived from digital elevation models (mean elevation, surface slope, and Topographic Position Index) and a capacity-distance-weighted Industrial Exposure Score (IES). A Random Forest regressor trained on this enriched 47-dimensional feature set achieves  $R^2 = 0.91$ , RMSE = 4.82, and MAE = 3.56, substantially outperforming an Ordinary Least Squares baseline ( $R^2 = 0.61$ ). Step-wise ablation experiments confirm that each feature group makes an independent contribution, with lagged exposure variables alone accounting for a  $\Delta R^2$  gain of +0.19. The resulting framework provides a scalable, operationally viable tool to support hospital resource planning and public health decision-making in pollution-burdened urban settings.

**Index Terms**—Air Quality Index (AQI), Respiratory Hospital Admissions, Lag Features, Random Forest, Topographic Analysis, Industrial Exposure Score, India, Predictive Modeling, Public Health.

## I. Introduction

Ambient air pollution has emerged as one of the most consequential environmental health challenges of our era, responsible for an estimated 7 million preventable deaths per year across the world [2]. The respiratory tract, as the primary anatomical gateway through which airborne contaminants enter the human body, sustains the sharpest immediate impact—manifesting as asthma exacerbations, chronic obstructive pulmonary disease (COPD), acute bronchitis, and lower respiratory infections [1]. Fine particulate matter (PM<sub>2.5</sub>), coarse particles (PM<sub>10</sub>), nitrogen dioxide (NO<sub>2</sub>), and sulfur dioxide (SO<sub>2</sub>) are among the most studied drivers of this burden, each independently associated with increased hospitalization risk [3].

India offers a particularly compelling context in which to study these dynamics. More than 300 of its urban centers routinely exceed both national regulatory thresholds and WHO ambient air quality guidelines, placing India near the top of global rankings for urban pollution burden [10]. Rapid industrial expansion, heavy reliance on coal-generated electricity, high vehicular density, and sharply varying topography combine to produce pollution patterns that diverge substantially across cities and seasons—rendering generic, spatially undifferentiated modeling strategies inadequate for capturing real-world exposure complexity.

A critical complication that continues to be underappreciated in predictive modeling is the temporal gap separating pollutant inhalation from observable clinical deterioration. Respiratory pathology does not emerge instantaneously; inflammatory cascades, oxidative tissue injury, and dysregulated immune responses typically require days to reach the clinical threshold of hospital presentation [4]. Although epidemiological time-series literature has thoroughly characterized this lag structure [5], the vast majority of data-driven admission prediction systems model the pollution-to-hospital link as if no delay exists—a simplification that systematically underestimates health impacts [6].

Geographic setting introduces an additional layer of complexity. Cities situated in low-lying terrain enclosed by higher ground accumulate pollutants more readily because topographic barriers suppress atmospheric mixing and ventilation [9]. Conversely, elevated or topographically open cities benefit from stronger dilution capacity. Beyond terrain, large fossil-fuel power installations—especially coal-fired thermal plants—superimpose a persistent background layer of pollution that keeps baseline exposure elevated even on days when episodic readings appear moderate [12].

To our knowledge, no existing modeling framework simultaneously addresses: (i) the multi-day lag structure governing pollutant health effects; (ii) cross-city geographic variability across a large, heterogeneous national sample; (iii) DEM-derived terrain characteristics as local exposure modifiers; and (iv) industrial point-source proximity as a chronic background covariate. The present study addresses this gap through four principal contributions:

- 1)** Construction of a seven-day lag feature set drawn from both composite AQI and four constituent pollutant time series, specifically designed to encode the delayed biological response to pollution episodes.
- 2)** Derivation of three city-level terrain descriptors—mean elevation, surface slope, and Topographic Position Index—extracted from high-resolution SRTM digital elevation models.
- 3)** Development of a novel Industrial Exposure Score (IES) that integrates the nameplate generating capacity and geographic proximity of fossil-fuel power stations to quantify chronic industrial pollution pressure on each city.
- 4)** Training and systematic evaluation of a Random Forest regression model across approximately 300 Indian cities, with rigorous comparison against an Ordinary Least Squares baseline.

## **II. Related Work**

### **A. Air Pollution and Respiratory Health**

The epidemiological foundation linking ambient particulate matter to adverse health outcomes was established by Pope and Dockery [1], whose longitudinal cohort studies documented a clear dose-response relationship between long-term PM<sub>2.5</sub> exposure and all-cause mortality. Brunekreef and Holgate [4] subsequently traced the biological mechanisms to systemic inflammation and oxidative cellular damage, providing a physiological rationale for these associations. WHO guidelines [2] synthesize this evidence to assign a substantial fraction of the global respiratory burden to ambient air pollution. Dominici et al. [3] contributed county-level evidence from a large U.S. Medicare cohort demonstrating acute-phase PM<sub>2.5</sub> elevations are reliably followed by spikes in respiratory admissions. At the planetary scale, Burnett et al. [11] estimated that outdoor pollution accounts for roughly 3.7 million deaths annually.

### **B. Time-Series and Lag Effect Studies**

Bell, Samet, and Dominici [5] synthesized the distributed lag structure of PM<sub>10</sub> and PM<sub>2.5</sub> effects across geographically diverse study sites, confirming robust multi-day associations between particulate exposure peaks and subsequent admission elevations. A key methodological implication of this work—that distributed lag specifications are necessary to capture the full temporal exposure-response curve—has rarely been applied in machine learning-based prediction systems. From the statistical learning perspective, Hastie, Tibshirani, and Friedman [6] highlight the advantages of ensemble tree methods over linear approaches when input-output relationships are non-linear and high-dimensional, as is characteristic of pollution-health data.

### **C. Spatial Analysis and Land-Use Regression**

Hoek et al. [8] conducted a comprehensive evaluation of land-use regression (LUR) models and identified road traffic density, land-cover type, and distance from emission sources as the strongest spatial predictors of local PM<sub>2.5</sub> variation. Jerrett et al. [9] demonstrated that within-city exposure gradients driven by these spatial factors translate into detectable differential mortality rates at fine geographic resolution. The role of terrain in shaping pollutant accumulation—particularly the tendency of enclosed valley topography to generate temperature inversions and trap

emissions—is acknowledged in the literature [10] but has seldom been operationalized at scale in multi-city regression frameworks.

#### **D. Industrial Exposure and Machine Learning**

Guttikunda and Gurjar [10] quantified the disproportionate footprint of coal-fired thermal power generation in India, estimating that such facilities account for approximately 30% of national SO<sub>2</sub> emissions—a contribution that saturates the local airshed for surrounding communities. The WRI Global Power Plant Database [12] provides the georeferenced, publicly accessible inventory needed to construct distance-weighted industrial exposure metrics at city scale. Breiman [7] established that Random Forest regression is highly resistant to overfitting and adept at modeling the non-linear, high-order interactions that characterize pollution-health relationships and that parametric linear models cannot represent. No prior study has unified lag-structured temporal features, terrain-based covariates, and industrial proximity scores within a single ensemble architecture applied to respiratory admission prediction across India.

### **III. Methodology**

#### **A. Data Collection**

Four distinct data streams were assembled. Daily air quality observations—comprising composite AQI alongside PM<sub>2.5</sub>, PM<sub>10</sub>, NO<sub>2</sub>, and SO<sub>2</sub> concentrations—were obtained for approximately 300 Indian cities from January 2020 through December 2025, drawing on the Central Pollution Control Board (CPCB) monitoring network and the OpenAQ API; city-day records where more than 20% of observations in any rolling 30-day window were missing were excluded before analysis. Terrain characteristics at 30-meter spatial resolution were extracted from Shuttle Radar Topography Mission (SRTM) GeoTIFF files to describe the elevation landscape surrounding each urban center. Power plant records were sourced from the WRI Global Power Plant Database [12], which provided geographic coordinates, fuel classification, and nameplate capacity for all fossil-fuel facilities within 200 km of each city centroid. Respiratory hospital admission counts were generated through a synthetic simulation procedure calibrated to published Indian hospitalization rate estimates, encoding realistic epidemiological dynamics including concurrent AQI levels, lagged exposure histories, city population, seasonal patterns, and stochastic variability.

#### **B. Data Preprocessing**

Missing values in the AQI and pollutant time series were recovered via a two-stage procedure that first applied seasonal-trend decomposition to model the underlying temporal structure and then filled residual gaps using k-nearest-neighbor (KNN) imputation—an approach that preserves autocorrelation rather than introducing artificial breaks. Observations deviating more than  $\pm 3$  interquartile range units from city-specific monthly medians were flagged as outliers and replaced through linear interpolation. To prevent information leakage from future observations into the training phase, z-score normalization parameters were derived exclusively from the training partition and subsequently applied across all data splits. After completing all quality-filtering and preprocessing steps, the consolidated dataset encompassed approximately 547,500 city-day records (roughly 300 cities  $\times$  1,825 days), with around 85% of observations meeting the quality inclusion criteria.

#### **C. Feature Engineering**

Feature construction was organized into three stages, each designed to capture a conceptually distinct dimension of the exposure-admission relationship:

**1) Lag Features:** For each city-day observation at time  $t$ , historical values of both composite AQI and each of the four constituent pollutants were retrieved at lag offsets spanning one to seven days:

$$\text{Lag\_P}(t, k) = P(t - k), \quad k = 1, 2, \dots, 7 \quad (1)$$

Applying this procedure to five pollutant series across seven lag steps yields 35 lagged predictors per observation in addition to the contemporaneous readings, operationalizing the distributed lag exposure structure described by Bell et al. [5].

**2) Topographic Features:** Each city was characterized by three time-invariant terrain descriptors extracted from SRTM data. The Topographic Position Index (TPI) quantifies a location's elevation relative to its immediate surroundings:

$$\text{TPI}_i = \text{Elev}_i - \text{mean}(\text{Elev\_neighborhood}_i) \quad (2)$$

Positive TPI values identify topographic highs (ridges and plateaus) associated with strong atmospheric ventilation, negative values correspond to enclosed basins prone to pollutant accumulation, and near-zero values characterize flat terrain. This index serves as a computationally inexpensive proxy for local atmospheric mixing capacity.

**3) Industrial Exposure Score (IES):** The IES integrates the influence of surrounding power facilities by weighting each plant's generating capacity inversely with its geodesic distance from the city centroid:

$$\text{IES}_i = \sum_j [\text{Capacity}_j / (\text{Distance}_{ij} + \epsilon)] \quad (3)$$

Here,  $\text{Capacity}_j$  is the nameplate generating capacity of facility  $j$  in megawatts,  $\text{Distance}_{ij}$  denotes the geodesic separation in kilometers between facility  $j$  and city  $i$ , and  $\epsilon = 1$  is a small regularization constant that prevents division by zero for co-located facilities. The resulting feature matrix comprises 47 dimensions per observation: five contemporaneous pollutant readings, 35 lagged values, three terrain descriptors, one IES term, and three temporal encoding variables.

#### D. Model Design

**1) Linear Regression (Baseline):** The OLS baseline estimates model coefficients by minimizing the total squared prediction error:

$$\min_{\beta} \|y - X\beta\|^2 \quad (4)$$

**2) Random Forest Regression:** Consistent with Breiman [7], the ensemble combines  $B = 200$  decision trees, each fitted to a bootstrapped subsample of the training data with  $m = \sqrt{47} \approx 7$  candidate features evaluated at each node split. Final predictions are obtained by averaging across all individual tree outputs:

$$\hat{y} = (1/B) \sum_{b=1}^B h_b(x) \quad (5)$$

Predictive performance is quantified using three complementary metrics:

$$\text{RMSE} = \sqrt{(1/n) \sum (y_i - \hat{y}_i)^2} \quad (6)$$

$$\text{MAE} = (1/n) \sum |\hat{y}_i - y_i| \quad (7)$$

$$R^2 = 1 - \sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2 \quad (8)$$

#### IV. Experimental Setup

All computations were implemented in Python 3.10, utilizing scikit-learn for machine learning operations, NumPy and pandas for numerical and tabular data management, and GeoPandas for spatial data handling. Terrain raster processing relied on GDAL and richDEM. Data were divided chronologically to prevent any temporal leakage: city-day records spanning January 2020 through December 2023 (73% of all observations) formed the training set, and the remaining 27% covering January 2024 through December 2025 constituted the held-out evaluation set. Hyperparameter tuning was conducted via RandomizedSearchCV employing a 5-fold TimeSeriesSplit cross-validation scheme. The optimal configuration identified through this procedure was:  $n\_estimators = 300$ ,  $max\_depth = 40$ ,  $min\_samples\_leaf = 4$ , and  $max\_features = 7$ . Full model training on an Intel Core i9-12900K workstation equipped with 64 GB of RAM required approximately 47 minutes.

**TABLE I**  
Dataset Summary Statistics

| Feature / Variable | Range / Values | Source |
|--------------------|----------------|--------|
|--------------------|----------------|--------|

|                                    |                |               |
|------------------------------------|----------------|---------------|
| AQI (t)                            | 0 – 500        | CPCB / OpenAQ |
| PM2.5 ( $\mu\text{g}/\text{m}^3$ ) | 5 – 450        | CPCB          |
| PM10 ( $\mu\text{g}/\text{m}^3$ )  | 10 – 600       | CPCB          |
| NO2 ( $\mu\text{g}/\text{m}^3$ )   | 2 – 180        | CPCB          |
| SO2 ( $\mu\text{g}/\text{m}^3$ )   | 1 – 120        | CPCB          |
| Lag AQI (t-1 ... t-7)              | 0 – 500 (each) | Derived       |
| Mean Elevation (m)                 | 8 – 2,400      | SRTM DEM      |
| Mean Slope ( $^\circ$ )            | 0.1 – 28       | SRTM DEM      |
| TPI                                | -180 – 220     | SRTM DEM      |
| Industrial Exposure Score          | 0 – 8,400      | WRI [12]      |
| Resp. Admissions (target)          | 0 – 320        | Synthetic     |

## V. Results and Analysis

### A. Overall Model Performance

Table II summarizes test-set evaluation results for both models. The Random Forest regressor outperforms the OLS baseline across every reported metric by a substantial margin. An  $R^2$  of 0.91 versus 0.61 represents a nearly 50% improvement in explained outcome variance, while the RMSE drop from 9.12 to 4.82 admissions per city-day—a 47% reduction—translates directly into more reliable operational forecasts. These gains align with the theoretical expectation that the relationship between pollutant exposure and hospital admissions is governed by non-linear, high-order interactions that ensemble tree architectures are equipped to learn but that ordinary linear regression structurally cannot represent.

**TABLE II**  
**Model Performance Comparison (Test Set)**

| Model                        | RMSE  | MAE   | $R^2$  |
|------------------------------|-------|-------|--------|
| Linear Regression (Baseline) | 9.12  | 7.43  | 0.61   |
| Random Forest (Proposed)     | 4.82  | 3.56  | 0.91   |
| Improvement (%)              | 47.1% | 52.1% | +49.2% |

### B. Ablation Study

Table III presents the step-wise ablation results. Beginning with contemporaneous AQI as the sole predictor, each successive feature group yields consistent and measurable accuracy improvements. Adding the seven-day lag history alone reduces RMSE by 33.5% and raises  $R^2$  from 0.65 to 0.84, establishing temporal exposure history as the single most impactful feature class. The subsequent inclusion of terrain descriptors produces an additional 7.5% RMSE reduction ( $R^2 = 0.88$ ), and appending the IES delivers a comparable further gain, bringing the full model to  $R^2 = 0.91$ .

**TABLE III**  
**Ablation Study – Feature Group Contributions**

| Feature Configuration        | RMSE | MAE  | R <sup>2</sup> |
|------------------------------|------|------|----------------|
| Contemporaneous AQI only     | 8.47 | 6.91 | 0.65           |
| + Lag features (t-1 ... t-7) | 5.63 | 4.22 | 0.84           |
| + Topographic features       | 5.21 | 3.89 | 0.88           |
| + Industrial Exposure Score  | 4.82 | 3.56 | 0.91           |

### C. Feature Importance Analysis

Permutation-based importance analysis surfaced several notable patterns. Three-day lagged AQI (Lag-3) ranked as the single strongest individual predictor, consistent with the 2–5 day window between peak particulate exposure and peak admission rates established in the epidemiological literature [5]. Short-lag AQI terms at offsets of one, two, four, and five days collectively occupied six of the ten highest-ranked positions, further affirming that recent exposure history carries substantially more predictive signal than contemporaneous readings. Among pollutant species, PM<sub>2.5</sub> ranked ahead of PM<sub>10</sub>, NO<sub>2</sub>, and SO<sub>2</sub>, mirroring existing evidence regarding its relative severity as a respiratory hazard [1][3]. The IES placed eighth in overall importance—well above terrain variables—reflecting the predictive value of chronic industrial exposure. Mean Elevation and TPI ranked 11th and 13th respectively, confirming that terrain descriptors supply statistically meaningful information that neither pollution concentrations nor industrial proximity alone can provide.

### D. Spatial and City-Level Performance

Geographic variation in prediction accuracy was both meaningful and interpretable. Cities within the Indo-Gangetic Plain—characterized by low-relief terrain, elevated IES values, and chronically high PM<sub>2.5</sub> concentrations—achieved the highest accuracy (mean R<sup>2</sup> = 0.94), benefiting from conditions that the feature set is well designed to represent. Coastal cities performed somewhat less accurately (mean R<sup>2</sup> = 0.87), with the shortfall attributed to marine boundary layer dynamics and sea-breeze circulation—processes that static terrain descriptors cannot adequately capture. Hill station cities fell in between (mean R<sup>2</sup> = 0.89), suggesting that DEM-derived elevation features successfully encode the ventilation advantages characteristic of elevated topographic settings. Cities situated within 50 km of coal plants with nameplate capacities exceeding 1,000 MW consistently recorded higher predicted admission baselines, providing empirical corroboration for the IES as a valid indicator of chronic background exposure [10].

## VI. Discussion

The prominence of three-day lagged AQI as the model’s leading predictor is consistent with a foundational principle of distributed lag epidemiology [5]: the full health toll of a pollution episode accumulates progressively over the days that follow, driven by the kinetics of pulmonary inflammation, oxidative injury, and impaired mucociliary clearance. This finding carries direct implications for the design of public health early warning systems. Most current alert protocols rely on same-day or previous-day AQI thresholds, which are structurally unable to anticipate the admission surges that materialize three to five days after an exposure peak. Incorporating the prior three-to-five-day pollution trajectory into alert logic could meaningfully extend the forecast horizon available to hospital administrators for proactive resource staging.

The predictive salience of the IES corroborates epidemiological documentation of the long-term health toll imposed by thermal power facilities in India [10][11]. A distinctive pattern observed in the city-level results is that communities within range of large coal plants tend to exhibit elevated admission baselines even on days when AQI readings are nominally moderate, pointing to cumulative toxicological burden rather than acute exposure as the operative mechanism. This observation strengthens the case for enhanced ambient monitoring and targeted health surveillance in communities near high-capacity industrial point sources, where the gap between official pollution readings and true physiological burden may be widest.

That terrain features contribute statistically independent predictive value is in line with LUR literature demonstrating that topographic structure governs local pollutant accumulation and dispersion [8]. The Topographic Position Index in particular provides an efficient, computationally inexpensive approximation of atmospheric mixing capacity that would otherwise require running high-resolution numerical weather prediction models—a practical advantage when the modeling objective is to scale analysis simultaneously across hundreds of cities. The relative performance gap for coastal cities ( $R^2 = 0.87$ ) reflects an inherent limitation of this approach: static terrain descriptors cannot represent the dynamic, wind-driven processes—including sea-breeze circulation, onshore flow regimes, and marine boundary layer height variability—that dominate pollutant dispersion in coastal environments.

From an operational standpoint, the framework supports forward-looking hospital resource management with admission forecasts carrying lead times of up to seven days. Such projections could enable public health administrators to pre-position essential respiratory therapeutics, coordinate emergency department staffing proactively, and issue targeted health advisories before anticipated surges materialize rather than in response to them. Integration of this model into the data infrastructure of India's National Clean Air Programme (NCAP) would provide a concrete pathway toward near-real-time deployment at national scale.

## **VII. Conclusion**

This study developed and evaluated a lag-aware predictive framework for respiratory hospital admissions across approximately 300 Indian cities. By combining seven-day histories of AQI and constituent pollutant concentrations with DEM-derived terrain variables (mean elevation, surface slope, and Topographic Position Index) and a capacity-distance-weighted Industrial Exposure Score drawn from the WRI Global Power Plant Database [12], a Random Forest regressor trained on the resulting 47-dimensional feature space achieved  $R^2 = 0.91$ , RMSE = 4.82, and MAE = 3.56 on a two-year chronological hold-out partition—substantially exceeding the performance of an OLS baseline ( $R^2 = 0.61$ ).

Step-wise ablation analysis confirmed that each feature group delivers independent predictive value, while permutation importance analysis identified three-day lagged AQI as the dominant individual predictor and PM<sub>2.5</sub>-based features as the most influential pollutant group—both findings in close alignment with the epidemiological evidence base [1][3][5]. Collectively, this work fills a previously unaddressed gap by unifying temporal lag dynamics, multi-city geographic heterogeneity, terrain-mediated pollutant accumulation, and industrial source proximity within a single cohesive predictive architecture.

## **VIII. Future Work**

Multiple directions merit investigation in subsequent research. Incorporating dynamic meteorological covariates—including wind speed and direction, atmospheric boundary layer height, and sea-breeze indicators—could address the performance deficit observed for coastal cities and improve accuracy in other wind-sensitive environments. Substituting the synthetic admission data with verified administrative records from Indian public health registries would substantially strengthen the clinical validity of the modeling framework and allow it to be benchmarked against real-world outcomes. Architecturally, sequence models designed for temporal learning—such as Long Short-Term Memory (LSTM) networks or Temporal Convolutional Networks (TCNs)—could uncover exposure-response dependencies extending well beyond the seven-day window examined here. Expanding the lag horizon to 14–21 days may prove particularly valuable for modeling slower-developing chronic conditions such as COPD or pulmonary fibrosis. On a longer time horizon, projecting future admission burdens under IPCC climate emission trajectories combined with urban growth forecasts could yield policy-relevant estimates of how evolving pollution patterns will reshape hospital demand over coming decades. Finally, applying causal inference methodologies could help isolate the attributable contribution of individual pollution sources and generate admission fraction estimates useful in cost-benefit analyses of targeted pollution control interventions.

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